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SECTION 1

Tips by the Five Layers

- **Persona layer:** Treat instructions as an API contract with a crisp role, explicit exclusions, repeated guardrails, a structured “I don’t know” path, and runtime facts injected sparingly to avoid token bloat.
- **Tools and actions:** Ship single-purpose tools with clear schemas and docstrings, control when tools may run, cap exposed MCP surface area, and return structured errors so the model can recover.
- **Reasoning and planning:** Use ReAct or plan-then-execute for multistep work, cap turns, add a brief self-review before final output, and dial reasoning down for one-shot tasks.
- **Knowledge and memory:** Make RAG a first-class tool, separate session thread context from long-term vector memory, hybridize search, chunk semantically, ground answers strictly in retrieval, and partition corpora by role or tenant.
- **Evaluation and feedback:** Trace prompts, tools, latency, and outcomes; automate eval suites and guardrails; capture human feedback; and iterate with an AIOps loop across indexes and APIs the agent touches.

SECTION 2

Customer Support Agent

- **Narrow charter:** Begin with bounded intents such as orders, returns, and status, expanding domains only after grounding and escalation paths prove reliable.
- **Grounding and safety:** Route answers through RAG over policies and manuals, add grounding checks before users see text, and verify identity before account or refund actions under least privilege.
- **Human escalation:** Expose an escalate tool for angry users, ambiguous policy, or high-impact refunds, and collect per-conversation feedback for later eval review.
- **Resilience and cost:** Apply retries, fallbacks, timeouts, caching of frequent policy answers, rate limits on external APIs, and transparent traces of searches, tools, and citations for audit.
- **Multi-agent workflow:** Production support is usually a chain—triage, retrieval, grounding, scoped API action behind a guardrail, final answer—not a single monolithic bot.

SECTION 3

RAG Agent System

- **Orchestrate only when needed:** A one-shot retrieve-and-answer path is enough for many cases; add agents when routing, critique, or post-retrieval actions justify the overhead.
- **Modular roles:** Split router or triage, retriever, answerer, and optionally a critic that triggers corrective retrieval when passages fail to support the question.
- **Retrieval quality:** Tune ANN indexes, metadata filters, hybrid search, reranking, and embeddings; log recall pain points and cases that should have ended in “I don’t know.”
- **Grounding discipline:** Instructions must use only retrieved context, require citations or excerpts, and prefer honest uncertainty over confident wrong answers.
- **Refinement loops:** When ranking shows weak coverage, a refinement agent may rewrite the query and search again behind a bounded retry counter to prevent infinite retrieval cycles.

SECTION 4

Deep Research Agent System

- **Two-tier orchestration:** A research planner breaks goals and delegates to stateless workers—web search, extraction, analysis, writing—while keeping long memory out of tool-like hands.
- **Tool policy:** Force factual claims through retrieval or web tools; forbid unsourced assertions in the planner’s instructions so synthesis stays tied to evidence.
- **Critique before delivery:** A critic pass checks coverage, bias, and contradictions; insufficient material returns control to the planner rather than shipping a thin brief.
- **Streaming experience:** Stream partial findings or status updates so users perceive progress and can approve checkpoints on long runs.
- **Budgets and cache:** Deduplicate repeated queries, cap tokens and external calls, and refresh caches on a schedule so speed does not trade away freshness.